

PROJECT TEAL

Two-Day User Testing Log & Iteration Report

Symptom Tracker Prototype

Researcher: Chunqing TangPlacement / Organisation: Papworth Trust Cambridge.Testing dates: 8.12Location: Cambridge PPW Site.Prototype version at start: 1.0 test prototype

Purpose

To quickly validate whether users can communicate a simulated symptom through the device and whether caregivers can correctly interpret the resulting data, then use the findings to make targeted design improvements between Day 1 and Day 2.

Scope: This is a lightweight usability and engineering validation exercise, not clinical validation. Use simulated scenarios only; do not induce pain or collect unnecessary medical details.

1. Testing Overview

1.1 Core validation questions

- Can a user understand when and how to use the symptom tracker with minimal explanation?
- Can the device distinguish a lighter vs stronger intended symptom input and a shorter vs longer event?
- Does the full hardware-to-app workflow record and synchronise events reliably?
- Can caregivers/staff interpret the recorded information quickly and correctly?
- What confusion, friction or unexpected behaviours appear in real use, and which are worth fixing immediately?

1.2 Success criteria for this two-day test

Measure	Target / acceptance criterion	Why it matters
Core task completion	Most users complete the core recording task with no more than one prompt	Basic <u>discoverability</u> and <u>usability</u>
Intensity ordering	For most users, <u>strong input > light input</u>	Shows the force input can encode <u>relative intensity</u>
Duration ordering	Long simulated event > short simulated event	Confirms duration capture works as intended
Bluetooth / app transfer	>= 90% of test events appear correctly in the app	End-to-end <u>system reliability</u>
Button confusion	No repeated systematic confusion between core functions	<u>Interaction clarity</u>
Caregiver interpretation	Most staff can identify strongest / longest / latest event correctly	Data is <u>useful</u> , not <u>merely collected</u>

1.3 Ethics, safety and privacy boundaries

- Use participant codes only (e.g. U01, U02, S01); do not write names in this document.
- Use imagined / simulated discomfort scenarios rather than asking participants to experience or disclose real pain.
- Follow Papworth Trust guidance on consent, capacity, photography and staff support.
- Stop a task if a participant appears uncomfortable, confused or wishes to stop.
- Keep staff interpretation separate from the participant's own response wherever possible.

1.4 Prototype configuration at the start of testing

Hardware version _____

Firmware version / date _____

App version / build _____

Functions enabled for testing ☒ symptom input ☒ haptic feedback ☒ recording ☒ BLE sync ☒ local storage ☐
other: _____

Known limitations before Day 1

*Recording Button Not Working
Visual que not clear on hardware.
Lose contact of hardware (prepare iteration).*

2. Day 1 - Formative Testing

Day 1 objective: Find the most important usability and system issues. Observe first; explain only when necessary. The goal is not to prove the prototype works perfectly, but to learn what to change.

2.1 Session flow (10-15 minutes per user)

Step	What to do / observe
1. Brief introduction	Explain that there are no right or wrong answers; you are testing the design, not the person.
2. Discoverability task	"Imagine something does not feel right. What would you do?" Observe the first action before prompting.
3. Intensity task	Ask for a "little uncomfortable" and a "very uncomfortable" simulated input; record sensor values.
4. Duration task	Ask for a short and a longer simulated event; check that recorded duration ordering is correct.
5. Optional secondary function	Only test recording / additional features if they are stable enough not to distract from the core task.
6. Quick feedback	Use a simple response such as Easy / Okay / Difficult, then ask "Was anything confusing?"

2.2 User session record

U05		Calibration		Not at all.		Easy		helpful. [But is it actually?]	
ID	Core task	Prompt s	Light FSR	Strong FSR	Duration correct?	Button confusion	BLE success	Ease	Keynote
U03	☒ I ☐ P ☐ N	Imagine Apain losing for 5s Apain gradually relieving	Also quite Strong	Too Strong It just saturated	☒ Y ☐ N	Not ingeneral. Explained once. (50s & Recording).	☒	☒ ☐ ☐ With	Happy about the haptic feedback. Duration precise. Pressure less Sensitive but is aware of the strength difference. Strong grip. FSR range too small
U04	☐ I ☐ P ☒ N	Imagine X press 5s x hard/gentle press v	No significant change in strength. Very gentle.	Don't understand long press Not used to haptic butt	☐ Y ☐ N	Turned off. slightly longer (25s).	☒	☐ ☐ ☒	Idea of time is vague. But might be better when experiencing actual pain.
U02	☒ I ☐ P ☐ N	Imagine hands if pain respond around	☒ 20%	☒ 90%	☒ Y ☐ N	Not at all	☒	☒ ☐ ☐	He understands it very well.
U01	☐ I ☒ P ☐ N	Press according to instruction	No significant change in strength of pressing But can tell when to press & when to let go	☐ Y ☒ N	Main Button fine. So instructed. Recording not attempted.	☒	☐ ☐ ☒	☐ ☐ ☒	Main Tracker button Can be identified without confusion. Don't tend to use So. Understands haptic

Legend: I = independent, P = prompted, N = not completed. BLE success = successful transfers / attempted transfers.

2.3 Staff / caregiver interpretation test

ID	Strongest event correct?	Longest event correct?	Latest event correct?	Approx. time	Realistic use case	Missing information	One change requested
S01	☒ Y ☐ N	☒ Y ☐ N	☒ Y ☐ N	☒	Calling for assistance		More training & clearer symptom selection.
S02	☒ Y ☐ N	☒ Y ☐ N	☒ Y ☐ N	☒	Signaling general discomfort.	understanding how the behaviour "pressing" is connected to feeling	Simpler interface.

2.4 Behavioural observations

Observation / what happened	Why it matters	Assumption challenged?	Possible design implication
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Feedback from Claire:

- ① Signs should be more graphic for better understanding
- ② Less words. Large graphs instead
- ③ tags: less options (no more than 3. Can set personalised common ones).
- ④ Notes: very difficult: ask parents for help.

Accessible Mode:
 Working document: use participant codes only; do not record identifiable health information.
 Almost do not use text at all.

Observation/what happened	Why it matters	Assumption challenged?	Possible design implication
Chris was not paying attention to the instruction/cannot imagine pain	Testing can be difficult	Look interesting & soothing enough for ppl to use.	
Calibration & actual reaction mismatch.	Calibration should be carefully instructed for feg precision.	Ppl don't automatically understand correspondence & purpose of calibration.	Add calibration prompts & visual que. for better understanding.
Rebecca doesn't understand "Imagine pain" or "If you have pain". But thinks it gives stomach ache.	When testing need to give clear instruction & simple explanation	Negative association with pain may cause reluctance to try the device.	more colourful look & softer tactile might help?

2.5 Day 1 immediate reflection

What surprised me most?

The test prompt "Imagine" is difficult for many.

Where did users hesitate or behave differently from what I expected?

Haptic feedback can cause unease

Which problem was caused by the interface, and which by the underlying concept?

Accessible mode mainly

What evidence suggests the current design is working?

What should NOT be changed yet because there is not enough evidence?

Haptic feedback.

3. Between-Day Iteration

Rule for overnight changes: Prioritise only critical and major issues. Do not add features just because someone suggested them. Every change should have evidence, a rationale and an expected effect that can be checked on Day 2.

3.1 Issue prioritisation

Issue	Evidence from Day 1	Severity	Decision	Reason
Accessible Mode Signs.	feedback.	☐ Critical ☐ Major ☒ Minor	☒ Change ☐ Defer ☐ Monitor	trivial & easy to change
BLE Transmission: repeated	multiple dots recorded for one event.	☒ Critical ☐ Major ☐ Minor	☒ Change ☐ Defer ☐ Monitor	Misreading of data.
Calibration Range.	observed during testing	☐ Critical ☒ Major ☐ Minor	☒ Change ☐ Defer ☐ Monitor	Affects only users with stronger grip.
Symbols added to the shell.	feedback.	☐ Critical ☐ Major ☒ Minor	☒ Change ☐ Defer ☐ Monitor	Better understanding.
		☐ Critical ☐ Major ☐ Minor	☐ Change ☐ Defer ☐ Monitor	
		☐ Critical ☐ Major ☐ Minor	☐ Change ☐ Defer ☐ Monitor	

3.2 Iteration log

Change made	Evidence / trigger	Engineering rationale	Expected effect on Day 2	Version / file reference

3.3 Updated prototype configuration for Day 2

Hardware changes Symbols added.

Firmware / logic changes X

App / UI changes larger, clearer symbols

Changes deliberately deferred _____

New version identifier _____

4. Day 2 – Confirmatory Testing

Day 2 objective: Repeat the same core tasks after targeted changes. The key question is whether the revised design improves performance or clarity, not whether every participant behaves identically.

4.1 Keep the test comparable

- Use the same core task wording and success criteria as Day 1.
- Prefer new participants where possible; if someone returns, mark them as a returning participant.
- Do not introduce extra explanation unless it would also be available in normal use.
- Record failures and unexpected behaviour exactly as carefully as successes.

4.2 User session record

ID	Core task	Prompt s	Light FSR	Strong FSR	Duration correct?	Button confusion	BLE success	Ease	Key note
U05	☒ I ☐ P ☐ N				☐ Y ☐ N	Much better	—/—	☒ 😊 ☐ 😐 ☐ 😞	
U__	☐ I ☐ P ☐ N				☐ Y ☐ N		—/—	☐ 😊 ☐ 😐 ☐ 😞	
U__	☐ I ☐ P ☐ N				☐ Y ☐ N		—/—	☐ 😊 ☐ 😐 ☐ 😞	
U__	☐ I ☐ P ☐ N				☐ Y ☐ N		—/—	☐ 😊 ☐ 😐 ☐ 😞	
U__	☐ I ☐ P ☐ N				☐ Y ☐ N		—/—	☐ 😊 ☐ 😐 ☐ 😞	
U__	☐ I ☐ P ☐ N				☐ Y ☐ N		—/—	☐ 😊 ☐ 😐 ☐ 😞	

4.3 Staff / caregiver interpretation test

ID	Strongest event correct?	Longest event correct?	Latest event correct?	Approx. time	Realistic use case	Missing information	One change requested
S01	☒ Y ☐ N	☒ Y ☐ N	☒ Y ☐ N				
S__	☐ Y ☐ N	☐ Y ☐ N	☐ Y ☐ N				

4.4 Behavioural observations

Observation / what happened	Improved vs Day 1?	What still failed or confused users?	Design implication
Claire's satisfied with changes in symbol,			
Chris is willing to try after detailed demo.			

Observation / what happened	Improved vs Day 1?	What still failed or confused users?	Design implication

4.5 Day 2 immediate reflection

Which change had the clearest positive effect?

Symbols.

Which issue remained unresolved?

Interpretation of the behaviour "Press" and how it corresponds to "pain"

Did any new problem appear because of the changes?

No.

What did the second day change my understanding of the user need?

Not much. Main needs were correctly identified.

What would I test next if I had another week?

AI trend Analysis / online model trained by known data,

5. Cross-Day Comparison&Conclusions

5.1 Quantitative / structured comparison

Metric	Day 1	Day 2	Change	Interpretation
Independent core-task completion				
Average prompts per user				
Intensity ordering success				
Duration ordering success				
Button-confusion rate				
BLE / app transfer success				
Caregiver interpretation accuracy				

5.2 Evidence-based conclusions

What can I reasonably claim the prototype can do after these two days?

What can I NOT claim yet?

Which design decision is now best supported by evidence?

What is the most important remaining uncertainty?

What is the next engineering step?

5.3 Suggested final one-sentence conclusion

Template: “Across two days of formative and confirmatory testing, the prototype [demonstrated / did not yet demonstrate] that users could _____ with _____ support, while caregivers were able to _____; the main remaining challenge is _____.”

6. Reflection, Leadership & LiA Evidence

6.1 Engineering reflection

Reflection prompt	Notes
A technical assumption that was wrong or incomplete	
A user behaviour that changed the design direction	
A trade-off I had to make (reliability, simplicity, functionality, time, manufacturability, etc.)	
An example of evidence changing a design decision	
What I would redesign from scratch with what I know now	

6.2 Leadership in Action reflection

Prompt	Reflection / evidence
Where did I take ownership or initiative?	
How did I adapt my communication to service users, caregivers or staff?	
What feedback did I actively seek, and how did I respond to it?	
How did I balance technical ambition with the needs and constraints of the organisation?	
What did this experience change about how I define responsible engineering leadership?	

6.3 SMART goal outcome summary

SMART goal	Evidence collected	Outcome / status	What changed because of the project?

6.4 Potential for wider / systemic impact

- Could the device or interaction model be integrated into existing support routines?

Working document - use participant codes only; do not record identifiable health information.

- What training, safeguarding, data governance or clinical involvement would be needed before wider use?
- Could the approach reduce reliance on retrospective verbal reporting?
- Which user groups or contexts might benefit, and which might not?
- What evidence would be required before scaling beyond a prototype?

7. Evidence Capture Checklist for Final Poster / Report

Evidence item	Captured?	File / location / note
Clear photo of prototype V1	☐ Yes ☐ No	
Photo / screenshot of app	☐ Yes ☐ No	
System architecture diagram (hardware - BLE - app)	☐ Yes ☐ No	
Photo of testing setup (only if permitted)	☐ Yes ☐ No	
Day 1 metrics table	☐ Yes ☐ No	
Day 1 key observation / quote	☐ Yes ☐ No	
Photo or note showing V1 -> V2 change	☐ Yes ☐ No	
Day 2 metrics table	☐ Yes ☐ No	
Cross-day comparison	☐ Yes ☐ No	
Staff / caregiver feedback	☐ Yes ☐ No	
One leadership reflection example	☐ Yes ☐ No	
One concise "next step / limitation" statement	☐ Yes ☐ No	

8. Final Summary (complete after Day 2)

Prototype version tested at the end _____

Number of service-user sessions _____

Number of staff / caregiver sessions _____

Most important validated finding _____

Most important unresolved issue _____

Immediate next step _____

Three-sentence project summary for the LiA poster/report:

1. Need / problem: _____

2. What I built and tested: _____

3. Evidence / outcome / next step: _____